

Solution to Exercise 4.6 (Page 54)

- Transformation of (4.49c) in standard form (4.44):

$$-x_1 + x_2 + x_3 + z = 1 \quad \text{with} \quad z \geq 0$$

- x_1 not included in (4.49d) (x_1 can in fact be negative) $\Rightarrow$ according to (4.47)

$$x_1 = x_1^+ - x_1^- \quad \text{with} \quad x_1^-, x_1^+ \geq 0$$

- Representation in standard form with $\bar{x} = [x_1^+, x_1^-, x_2, x_3, z]^T \in \mathbb{R}^5$

$$\begin{aligned} \min_{\bar{x} \in \mathbb{R}^5} & -x_1^+ + x_1^- + 2x_2 + x_3 \\ \text{s.t.} & \quad x_1^+ - x_1^- + x_3 = 1 \\ & \quad -x_1^+ + x_1^- + x_2 + x_3 + z = 1 \\ & \quad \bar{x}_i \geq 0, \quad i = 1, \dots, 5 \end{aligned}$$

Solution to Exercise 4.7 (Page 63)

- Linear optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^4} & \sum_{i=1}^4 K_i x_i \\ \text{s.t.} & \quad R_1 = x_1 + x_3 \quad (\text{consumption area } A_1) \\ & \quad R_2 = x_2 - x_3 + x_4 \quad (\text{consumption area } A_2) \\ & \quad P_1 \geq x_1 + x_2 \quad (\text{prod.cap. of } R_1) \\ & \quad 0 \leq x_i \leq C_i, \quad i = 1, \dots, 4 \quad (\text{cap. of pipeline } l_i) \end{aligned}$$

- Elimination of x_3 and x_4 by equality constraints

$$x_3 = R_1 - x_1, \quad x_4 = R_2 - x_2 + x_3 = R_1 + R_2 - x_1 - x_2$$

- Inequality constraints for x_3, x_4 :

$$\begin{aligned} 0 \leq \underbrace{R_1 - x_1}_{x_3} \leq C_3 : & \quad \boxed{x_1 \geq R_1 - C_3} \\ & \quad x_1 \leq R_1 \quad \text{already satisfied with } x_1 \leq C_1 < R_1 \\ 0 \leq \underbrace{R_1 + R_2 - x_1 - x_2}_{x_4} \leq C_4 : & \quad \boxed{x_1 + x_2 \geq R_1 + R_2 - C_4} \\ & \quad x_1 + x_2 \leq R_1 + R_2 \quad \text{satisfied with } x_{1,2} \leq C_{1,2} < R_{1,2} \end{aligned}$$

$\Rightarrow$ Reduced problem

$$\begin{aligned} \min_{x \in \mathbb{R}^2} & (K_1 - K_3 - K_4)x_1 + (K_2 - K_4)x_2 + \underbrace{K_3 R_1 + K_4(R_1 + R_2)}_{\text{negligible}} \\ \text{s.t.} & \quad x_1 + x_2 \leq P_1 \\ & \quad x_1 \geq R_1 - C_3 \quad (\text{from eq. constr.}) \\ & \quad x_1 + x_2 \geq R_1 + R_2 - C_4 \quad (\text{from eq. constr.}) \\ & \quad 0 \leq x_i \leq C_i, \quad i = 1, 2 \end{aligned}$$

Solution to Exercise 4.8 (Page 63)

- Problem with given parameters

$$\min_{\mathbf{x} \in \mathbb{R}^2} 20x_1 + 10x_2 \quad (\text{A.1a})$$

$$\text{s.t. } x_1 + x_2 \leq 45 \quad (\text{A.1b})$$

$$x_1 \geq 20, \quad x_1 + x_2 \geq 30 \quad (\text{A.1c})$$

$$0 \leq x_1 \leq 30, \quad 0 \leq x_2 \leq 20 \quad (\text{A.1d})$$

- Admissible set $\mathcal{X}_{ad}$, see Figure A.2

- Contour lines of cost fct.: $20x_1 + 10x_2 = m = \text{const.} \Rightarrow \boxed{x_2 = m - 2x_1}$

$\Rightarrow$ Optimal point $\mathbf{x}^* = [20, 10]^T$

Verification of sign condition (4.29d) for μ_i at $\mathbf{x}^*$:

- Cost fct. and inequality constr. (linear problem not in standard form)

$$f(\mathbf{x}) = 20x_1 + 10x_2, \quad h_1(\mathbf{x}) = 30 - x_1 - x_2 \leq 0, \quad h_2(\mathbf{x}) = 20 - x_1 \leq 0 \quad (\text{A.2})$$

- Gradients (see Figure A.2)

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 20 \\ 10 \end{bmatrix}, \quad \nabla h_1(\mathbf{x}) = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \quad \nabla h_2(\mathbf{x}) = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad (\text{A.3})$$

$\Rightarrow -\nabla f(\mathbf{x}^*)$ is positive linear combination of $\nabla h_1(\mathbf{x}^*)$ and $\nabla h_2(\mathbf{x}^*)$, i.e.

$$\nabla_{\mathbf{x}} L(\mathbf{x}^*, \boldsymbol{\mu}^*) = \nabla f(\mathbf{x}^*) + \mu_1^* \nabla h_1(\mathbf{x}^*) + \mu_2^* \nabla h_2(\mathbf{x}^*) = \mathbf{0}, \quad \boldsymbol{\mu}^* = [\mu_1^*, \mu_2^*]^T > \mathbf{0}$$

$\Rightarrow$ KKT conditions are satisfied

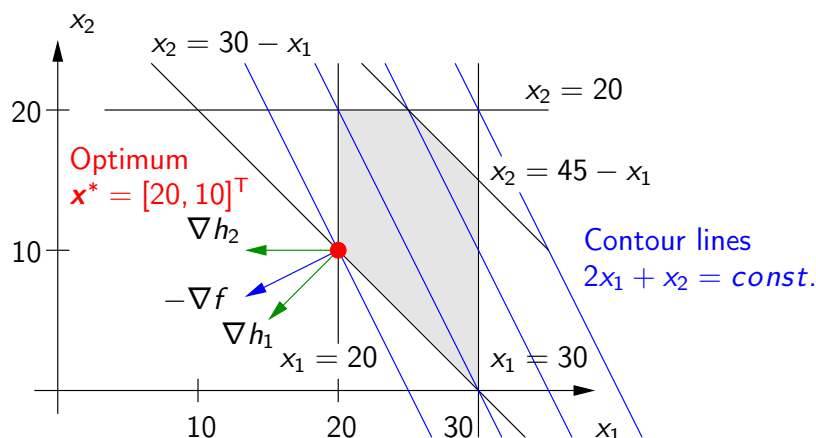


Figure A.2: Graphical solution of Exercise 4.7 (Gradients not true to scale!).